Tejal Nair

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Education

College of William & Mary — Williamsburg, VA
Ph.D. Student in Computer Science — Aug. 2025 – current

Cornell University — Ithaca, NY
Masters of Engineering in Computer Science — Aug. 2024 – May 2025
Bachelor of Arts in Mathematics and Computer Science — Aug. 2021 – May 2024

Experience

Professor Yanfu Zhang Lab - College of William & Mary — Aug. 2025 – current
Graduate Researcher

- Research Interests: LLMs, learning, AI for biomedicine
- Past works on demonstration ordering and retrieval for in-context learning and LLM trustworthiness
- Investigating how LLMs track and resolve conflicting information as state changes across multi-turn interactions and model updates, and developing uncertainty quantification methods to detect when their performance degrades.
- Investigating applications of in-context learning in biomedical domains, such as in protein structure.

Computer Vision Lab – NASA Langley Research Center
OSTEM Intern — June. 2026 – Aug. 2026

- Designed a spatio-temporal graph convolutional network (ST-GCN) on frozen DINOv2 features to classify aerial objects (birds, small UAS, planes) from multi-camera NASA flight-test video.
- Annotated and labeled past flights for aerial classification.

AMA Intern — Sep. 2026 – current

- Continued development of ST-GCN model for aerial object classification and detection.
- Writing paper on ongoing work for CVPR 2026 submission in November.

Modern Programming Fundamentals – College of William & Mary — Jan. 2026 – May 2026; Aug. 2026 – current
Graduate Teaching Assistant

- Taught and lead weekly lab sections.

Professor David Bindel Lab - Cornell University — Sep. 2022 – Aug. 2026
Rawlings Presidential Scholar/Undergraduate and Graduate Researcher

- Developed Gaussian Process Upper Confidence Bound (GP-UCB) framework incorporating inequality constraints and RKHS norm bounds for constrained function estimation and optimization.
- Implemented custom active-set QP solvers in Python and JAX, leveraging Cholesky factorization, KKT systems, and numerical linear algebra to solve constrained optimization problems.

Fundamentals of AI/ML – College of William & Mary — Aug. 2025 – Dec. 2025
Graduate Teaching Assistant

Introduction to Natural Language Processing – Cornell University — Jan. 2025 – May 2025
Graduate Teaching Assistant

Professor Bharath Hariharan Lab - Cornell University — Sep. 2024 – Dec. 2024
Graduate Researcher

- Explored the idea of using ML to speed up, rather than automate, annotation
- Clustered unlabeled bird data in a learnt feature space

Dr. Avi Ma'ayan Lab – Icahn School of Medicine at Mount Sinai — June 2024 – Aug. 2024
Bioinformatics Summer Intern

- Developed a website using Flask to analyze gene expression changes with age in different tissue types
- Developed both front-end and back-end web application to analyze RNA-seq data

Computer Systems Organization and Programming - Cornell University Aug. 2023 – May 2024
Undergraduate Teaching Assistant

- Mentored Middle School students on the basics of programming in Python

Dr. Jeffrey Chuang Lab - The Jackson Laboratory Feb. 2020 – Dec. 2020
High School Student Intern

- Worked on supervised-learning project classifying lung-cancer subtypes with CNNs using Tensorflow
- Presented poster at 2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)

Publications

Publications in reversed chronological order.

- **Towards Reliable, Generalizable, and Specific In-Context Knowledge Editing via Multi-Objective Reinforcement Learning**
  Xuzhong Wang, Maiqi Jiang, **Tejal Nair**, Girija Bhusal, Yanfu Zhang, Haipeng Chen
  EMNLP findings 2026

- **Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing**
  Tejal Nair, Mahmud Wasif Nafee, Maiqi Jiang, Ye Gao, Haipeng Chen, Yanfu Zhang
  ACL findings 2026

- **The effect of blurring on lung cancer subtype classification accuracy of convolutional neural networks**
  Tejal Nair, Ali Foroughi pour, Jeffrey H. Chuang
  IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2020

Papers Under Review

- **Proximity, Precendence and Position: A Minimum Linear Arrangement Account of Ordering Effects in LLM Prompts**
  Tejal Nair, Maiqi Jiang, Xuzhong Wang, Guodong Liu, De Wang, Qun Li, Haipeng Chen, Yanfu Zhang

- **Quantifying Graph Structural Memorization in Large Language Models**
  Zhenning Lu, Xuzhong Wang, Chenjie Ni, **Tejal Nair**, Ting Hua, Shangqian Gao, Qun Li, De Wang, Wei Chen, Guodong Liu, Yanfu Zhang

- **Representation Change and Adaptive Stopping in In-Context Learning**
  Tejal Nair*, Omid Mersa*, Zhen Xu*, Maiqi Jiang, Xuzhong Wang, Ashley Gao, Gang Zhou, Haipeng Chen, Yanfu Zhang

(* denotes equal contribution)